

Fraud Detection in Online Transactions

Project Description:

Company: PaySphere Digital Payments Pvt. Ltd.

Industry: FinTech, Digital Payments, Online Commerce Risk Management

PaySphere is one of India's fastest-growing digital transaction networks, processing over 65 million monthly payments across UPI, debit/credit cards, net banking, and stored wallets. The business operates at massive scale, powering transactions for ecommerce platforms, travel portals, gaming apps, and utility services.

Over the last quarter, PaySphere's Fraud Risk & Compliance Division has raised concerns about rising transaction anomalies and financial exposure. With the increased sophistication of fraudsters—using techniques such as device spoofing, account takeover, bot-based credential attacks, and rapid-fire payment retries—the company has observed:

- A 38% rise in attempted fraudulent transactions
- A marked increase in false negatives (fraud not caught in time) causing chargebacks
- A growing number of false positives, incorrectly blocking genuine customers and hurting platform revenue
- Significant class imbalance: fraudulent transactions form less than 0.5% of the total volume
- Difficulty in distinguishing unusual but genuine customer behaviour from malicious activity

The Chief Risk Officer has mandated a machine learning-driven fraud detection pipeline capable of real-time scoring. The intent is not merely to "flag bad transactions," but to design an intelligent risk engine that can learn patterns of fraud, reduce financial leakage, and protect customer trust.

The mandate includes:

- Identifying transaction-level signals associated with fraud
- Engineering behaviour-based features (frequency, velocity, transaction diversity)
- Handling severe class imbalance through resampling or algorithmic techniques
- Training classification models to maximise detection while controlling false alarms
- Performing threshold tuning to find the optimal fraud cut-off
- Providing a business-ready fraud risk scoring mechanism

- Recommending operational actions (hard block, soft review, OTP challenge, device lock)

The project reflects real-world challenges faced in financial fraud analytics—where risk, revenue, customer experience, and regulatory compliance intersect.

Stakeholders include:

- **Fraud Risk & Compliance Team** – accountable for reducing fraud losses and regulatory adherence
- **Payments Engineering Team** – responsible for integrating real-time scoring into the transaction gateway
- **Data Science & AI Team** – designers of the fraud detection models
- **Customer Experience Team** – focused on reducing false positives to avoid blocking genuine customers
- **Finance & Chargeback Team** – owners of fraud loss reporting, chargeback reconciliation, and merchant communication

A successful project delivers tangible uplift in:

- Fraud capture rate (True Positive Rate)
- Reduction in financial loss (chargebacks avoided)
- Balance between fraud blocked vs. legitimate user friction
- Operational efficiency for fraud analysts

PaySphere needs a predictive fraud detection model capable of distinguishing suspicious transactions from normal ones—based on patterns in transaction amount, customer behaviour, device consistency, merchant characteristics, geolocation, authentication methods, and temporal activity.

The challenge is heightened because:

- Fraud patterns evolve rapidly
- Genuine behaviour can sometimes resemble fraudulent behaviour
- Fraudulent transactions are extremely rare (highly imbalanced)

Your task is to design a robust ML-based fraud classification system that can be deployed in the transaction approval workflow.

Learners must validate the dataset, perform exploratory fraud analysis, engineer meaningful behavioural and risk-based features, and build a machine learning model to classify fraudulent transactions. They must evaluate model performance,

tune thresholds, and present clear business recommendations for fraud mitigation. A concise business report summarizing insights, model rationale, and operational actions is required.

Data Definition:

Use the file [**Data Definition.xlsx**] to understand fields, key relationships, and business rules.

Instruction to Learners:

Validate Data Integrity

Check all datasets for:

- Primary & foreign key consistency
- Null or incorrect values
- Logical/business rule violations
- Date correctness and chronological order. Make corrections wherever required.

Perform Feature Engineering

Create additional features needed for fraud analysis and classification (e.g., transaction velocity, behavioural deviation metrics, device/location consistency indicators, risk flags, and time-of-day or day-of-week features).

Suggested Feature Engineering Guide (For Reference Only)

This section provides business context behind a set of commonly used engineered features in fraud-detection systems. The guide is illustrative and helps learners understand how different behavioural, transactional, and risk-related variables may influence fraudulent activity patterns.

1. Customer & Behavioural Features

Transaction Velocity

Represents how many transactions a customer performs within a short time window (e.g., last 5 minutes, last hour).

Fraudsters often execute multiple rapid transactions before a cardholder notices or before a fraud rule blocks the card.

Average Transaction Amount (Historical)

Indicates typical spending behaviour for a customer.

Fraud amounts usually deviate — sometimes sharply — from a customer's usual transaction profile.

Amount Deviation Score

Captures how unusual the current transaction amount is compared to a customer's normal range.

Large deviations often trigger risk flags.

2. Device & Identity Consistency Features

Device Familiarity Score

Represents how often the customer has used this device in the past.

Fraud attempts often originate from **new** or **rarely seen** devices.

Account–Device Matching Indicator

Captures whether the customer has previously transacted with this device–IP–browser combination.

New combinations increase fraud likelihood.

IP Geolocation Distance

Measures geographical distance between customer's registered location and transaction IP location.

Large unexpected jumps — such as sudden overseas access — can signal account takeover.

3. Merchant & Category Risk Features

Merchant Risk Score

Summarises historical fraud exposure at the merchant level.

Categories such as digital goods, gaming, travel, or gift cards tend to attract higher fraud attempts due to instant fulfilment or resale value.

Merchant Consistency Indicator

Checks whether this customer has transacted with the merchant before.

Fraud transactions often target *new* merchants unfamiliar to the user.

4. Time-Based Fraud Indicators

Time-of-Day Risk

Represents behavioural patterns associated with unusual hours.

Fraud attempts spike during late-night hours when customers are less likely to monitor notifications.

Day-of-Week Risk

Fraud rings often exploit weekend periods when monitoring teams are thinner and banks may have slower manual review cycles.

5. Payment Method & Transaction Pattern Signals

High-Risk Payment Method Flag

Indicates whether the transaction uses methods historically associated with fraud spikes (e.g., international cards, virtual cards, one-click payments).

Amount Buckets

Groups transactions into low / medium / high ranges.

Certain fraud types involve small "test transactions" followed by larger fraudulent charges.

6. Network & Historical Fraud Indicators

IP Address Risk Score

Aggregates risk from regional fraud hotspots, known bad IPs, or anonymized proxy usage. Fraud attempts often originate from VPNs, TOR nodes, or data centers.

Historical Fraud Rate (Customer or Device)

Represents the proportion of past fraudulent transactions associated with the entity. Helps identify compromised accounts or repeatedly exploited devices.

7. Anomaly & Deviation Scores

Behavioural Anomaly Score

Measures how unusual the current transaction is across multiple dimensions (amount, velocity, device, location).

Fraud models rely heavily on anomaly detection because fraudsters constantly adapt to rules.

Combined Risk Index

A weighted score that merges signals from payment risk, device anomalies, merchant behaviour, and network exposure.

Provides a high-level fraud propensity estimate for downstream ML models.